AI-Powered Brain Tumor Detection Platform with Global Trend Mining and Geospatial User Analytics

Teammates Names:

Ishika Kaur

K Praveen Kumar

Karthik Idikuda

Problem Statement:

Brain tumors show varying patterns worldwide due to regional and environmental factors. Existing AI tools classify tumors from MRI scans but do not provide global context or user insights. This project creates a web platform that:

- Uses AI for real-time tumor classification from uploaded MRI scans (top priority).
- Incorporates Web Mining to extract current global brain tumor trends.
- Incorporates Web Analysis to track user locations and interactions. The aim is to show correlations between platform usage and global patterns for improved awareness.

Mining:

The project includes Web Mining to pull current global brain tumor trends from public sources.

- Display up-to-date incidence and mortality data by region/country.
- Include recent mentions of tumor clusters or environmental factors. All from free, legal public APIs and tools.

Web Analysis:

The project includes Web Analysis to monitor real user activity.

- Track anonymized interactions (uploads, views).
- Map user locations via IP geolocation (with consent).
- Display heatmap and compare with mined global trends.

Uniqueness of Project

- Integrates real-time AI tumor classification with Web Mining and Web Analysis.
- Correlates user detections and locations with global public trends.
- Provides Web Intelligence context not found in typical medical AI tools.

Outcome of Project

- **General Outcome**: Working web app for tumor detection plus interactive dashboard showing global trends and user patterns.
- **Future Use**: Aids awareness, research on regional factors, and public health monitoring.

Tech Stack

- **Programming Languages**
  - Python (backend & AI)
  - JavaScript (frontend)
- **Backend Framework**
  - FastAPI/Flask/ Django
- **Frontend Framework**
  - React.js, Vue.js, Plain HTML/CSS/JS + Bootstrap
- **Database**
  - PostgreSQL/MongoDB/SQLite/MySQL
- **AI/ML Libraries**
  - TensorFlow / Keras/PyTorch/scikit-learn + CNN
- **Maps & Geospatial Visualization**
  - Leaflet.js/Google Maps API/Folium

Modules :

1. Upload Module
2. AI Classification Module
3. Web Mining Module
4. Web Analytics Module
5. Dashboard Module

Conclusion :

The proposed platform represents a significant advancement over existing diagnostic tools by fusing the analytical power of AI with the contextual richness of global public health data. It moves beyond a simple, isolated classification to offer a multi-dimensional view of brain tumors. The intended outcome is a user-friendly web application that empowers individuals with contextualised knowledge, aids researchers in identifying new patterns, and provides public health officials with a novel lens for understanding the prevalence of brain tumors worldwide.